

**IE University**

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# **Ephemeral Interfaces: Design Patterns for Task-Scoped Generative UI Components**

*Subtitle of the project*

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**Abstract**

Your abstract goes here

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# 1 Introduction

How can ephemeral, task-scoped generative UI components be designed to improve existing applications without disrupting the user experience?

## 1.1 Literature Review

While LLMs have been primarily used to generate content through text, they are increasingly being used to generate user interfaces (UI). It is a way to move beyond static interfaces by generating structures based on a set of primitives tailored to the users' goals. Prior work suggests that when working with LLMs, users prefer to express themselves through UI instead of a plain chat. This is because the system can present its functionality properly instead of forcing it through text (**Leviathan2025GenerativeUI**; **Tang2025SurveyGUI**; **Ahmed2025LLMUIUX**). However, most of the evidence comes from the context of a fully generated UI, which differs from the design challenge of the capstone, which is to augment existing applications where users expect a certain degree of stability and predictability in their tools. Thus, this research motivates me to understand how we can use a set of core primitive UI components in already established workflows instead of replacing the whole state of our modern day interfaces with AI.

The problem with generative UI interfaces is that while they can increase the relevance of the task at hand, they can threaten predictability and user control. In real-world domains such as banking, their rigid interfaces were built on a set of predefined rules that fail to meet the ever-changing needs of their users across services (**Tambi2020GenerativeBanking**). Approaches that dynamically generate interfaces can address the problem of not tailoring just to the content needed to serve the user, but the actual structure of the interaction over time. Yet, these new systems raise an unresolved question: when generative UI is beneficial, how can it be introduced to users without it corroding the user's mental model of how the application should work? This implies that the design pattern put in place for generative UI must manage the tradeoff between adaptivity and consistency (**Bieniek2024GenerativeAI**).

To actually manage the tradeoff of leveraging generative UI, instead of having the entire

interface be generated with AI, we could restrict it to ephemeral, so that task-scoped components appear only when needed and disappear when their use case has passed. Past work on ephemeral interfaces argues that the permanence in ubiquitous environments can contribute to cognitive overload as there are too many controls and artifacts present (**Doring2013EphemeralUI**). Conversely, ephemerality can reduce the persistent clutter of artifacts and help with better interactions for the user within a specific time frame and context. While this research was grounded in physical interaction design, it provides principles that can be transferred to digital systems. These ephemeral elements should be discoverable when needed, easy to dismiss, and designed to disappear without affecting the surrounding environment's knowledge. For generative UI, it suggests that ephemeral components could be a natural fit for LLMs as the system could generate the UI contextually in almost real time (**Bieniek2024GenerativeAI**). However, it must be kept in mind how the component lifecycle should be carefully designed to prevent an unstable user experience.

Ephemeral generative components have been implemented in past research without a full interface replacement, but with a temporary UI as an intermediate layer for exploration and understanding. For agentic programming, research used ephemeral interfaces that sat between the prompt and code generation to improve observability and comprehension of LLM generated code. It helped developers steer their results without needing to inspect the raw code (**Cheng2024Biscuit**). This shows how ephemeral UI can function as a way to help users with a workflow that lets them steer the outcome and iteratively refine stochastic tasks for exploring alternatives before taking concrete actions. However, this application of ephemeral UI was done in a narrow domain and intent. This leaves the question of how ephemeral patterns can generalise to everyday productivity applications where tasks are diverse. If not implemented properly it could result in a disruption of interruptions, context switching, or inconsistency of familiar workflow.

Taking all existing literature into account, it establishes three relevant and incomplete variables to this capstone. Firstly, fully generative UI experiences can improve the user experience and enjoyment of interacting with generative AI applications compared to only chat-only interaction (**Leviathan2025GenerativeUI**; **Tang2025SurveyGUI**; **Ahmed2025LLMUIUX**).

Secondly, deployments of dynamic interface generations in specific domains can address the shortcomings of problems in need of rule-based logic where needs are constantly changing and require tailoring (**Tambi2020GenerativeBanking**). Thirdly, research on ephemerality shows how temporary time-bound interaction artifacts can reduce cognitive overload by decreasing the amount of permanent UI accumulation for the user. Nonetheless, a clear gap remains at the intersection of limited guidance on how to design *ephemeral, task-scoped generative UI components* that can improve existing applications while preserving a predictable user experience. Prior work does not articulate reproducible design patterns that define when generative components should appear, how their scope should be bounded with an exit that does not disrupt the users' mental model of the application, and how they should preserve a predictable user interface. This capstone addresses that gap by generalising ephemerality into a framework of component patterns to integrate into everyday applications.

## 2 Methodology

### 2.1 Research Design

This project adopts a Design Science Research (DSR) approach because its primary goal is not only to study an existing phenomenon, but to create and evaluate an artifact that addresses an identified design problem. The problem established in the literature is that current generative interfaces often fall at one of two extremes: either assistance is constrained to static interface structures, or entire experiences are dynamically generated in ways that can undermine predictability and continuity. In response, this project proposes a conceptual framework for ephemeral, task-scoped generative UI components intended to augment existing applications without disrupting user workflow. The methodological contribution therefore lies in developing a framework that translates ideas from prior work on ephemerality, generative UI, and interaction support into a form that can guide design decisions in a broader productivity context.

Following this DSR logic, the study proceeds in three connected stages. First, a conceptual framework is synthesised from the literature in order to define the taxonomy, lifecycle, and design principles of ephemeral interface components. Second, that framework is instantiated in a website artifact that serves as a research instrument, allowing the proposed ideas to be embodied in a controlled interactive environment. Third, the artifact is evaluated through a comparative user study that contrasts an original interface with an ephemeral condition on a single, bounded task. This sequence ensures that the evaluation is not detached from theory: the framework informs the design of the artifact, and the artifact in turn provides the basis for assessing whether ephemeral augmentation can improve task performance and user experience without introducing excessive intrusiveness.

### 2.2 Conceptual Framework

This study develops a conceptual framework for ephemeral, task-scoped generative UI components to guide both artifact design and evaluation. In one sentence, the framework specifies *which temporary support roles may appear, how those supports are triggered and withdrawn,*

and *which constraints keep them useful without destabilising the host interface*. Positioning the framework here in the methodology is important because it defines the concepts that the artifact operationalises and the criteria against which the two interface conditions are later compared.

The first element is a taxonomy of component roles. Rather than classifying components by visual form, such as whether they appear as a panel, popover, or inline element, the taxonomy classifies them by the function they serve within the workflow. Three roles are distinguished. *Interpretive support* helps users understand context, available options, outputs, or likely consequences before acting. *Refinement support* helps users steer or configure an intended action before commitment, for example by comparing alternatives or adjusting parameters. *Task-execution support* helps users complete a bounded subtask inside the surrounding application without replacing the broader workflow. Framing these roles as a taxonomy is methodologically useful because it makes explicit what kind of assistance is being introduced in each scenario and avoids treating ephemeral behaviour as a single undifferentiated interface style.

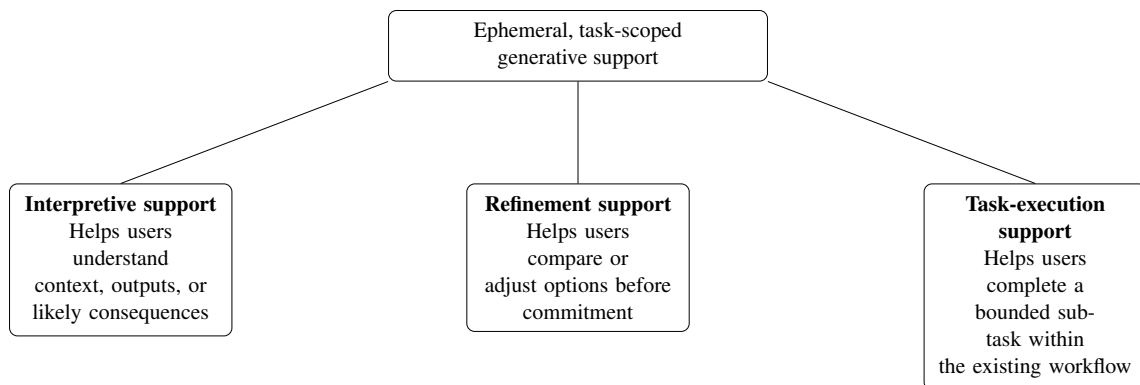


Figure 1: Taxonomy of ephemeral, task-scoped generative support roles.

The second element is a lifecycle describing how an ephemeral component enters and leaves use. Alternative stage patterns can be found in the literature, ranging from the minimal logic of temporary appearance and disappearance to more scaffolded models in which users configure support before committing to an outcome. For this study, the most appropriate synthesis is a four-stage lifecycle of *trigger*, *instantiation*, *interaction*, and *dismissal*. A component is triggered when a clear user goal or contextual need is detected that is not adequately supported by the persistent interface alone. That trigger may be explicit, such as a user request for help, or inferred from local behavioural and contextual signals, but only when the resulting sup-

port remains low-risk, relevant, and easy to dismiss. The component is then instantiated as a temporary interface element embedded within the existing application rather than replacing the surrounding interface. During interaction, users inspect, steer, or complete the immediate subtask while remaining oriented within the primary workflow. Finally, the component is dismissed once the task is completed, abandoned, or no longer contextually relevant. This lifecycle was selected because it retains the conceptual clarity of ephemerality while remaining concrete enough to implement as a state-based interaction model in the artifact.

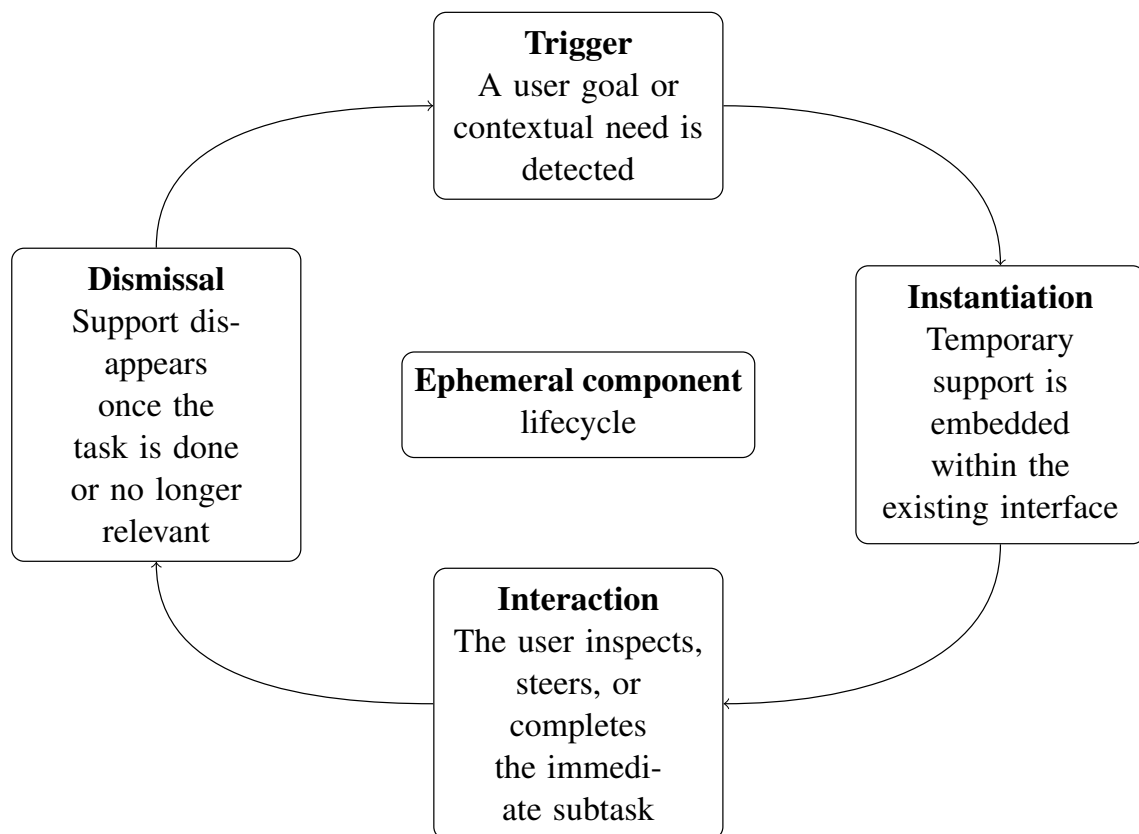


Figure 2: Lifecycle of an ephemeral, task-scoped generative UI component: trigger, instantiation, interaction, and dismissal.

The third element is a set of design principles that constrain how ephemeral components should be introduced into an otherwise stable interface. Five principles guide the framework. *Contextual relevance* means support should appear only when it serves a clear task-related purpose in the current situation. *Bounded scope* means support should remain limited to the immediate subtask rather than taking over the larger workflow. *Interface continuity* means the temporary component should preserve the surrounding page structure and the user's sense of place, rather than redirecting the interface toward an unrelated experience. *User control* means



users should be able to inspect, refine, confirm, or dismiss the support rather than being forced into a system decision. *Graceful disappearance* means the component should recede cleanly once it is no longer needed. When support is inferred rather than explicitly invoked, these principles impose an additional safety constraint: inferred assistance may suggest or preload options, but it should not perform destructive or strongly consequential actions without explicit user confirmation. Together, these principles translate the theoretical idea of ephemerality into practical design criteria that can be applied consistently during artifact development and later assessed in the evaluation.

## 2.3 Artifact Development

The artifact developed in this project is a website used as a research instrument for evaluating the proposed framework in a controlled but realistic interaction setting. Rather than building a fully autonomous or fully generative application, the website is designed to preserve a recognisable baseline interface while allowing ephemeral, task-scoped components to be introduced at selected moments. This makes it possible to test whether the framework can augment an existing user experience without replacing the primary workflow altogether.

To operationalise the framework, the website will implement two interface conditions: an original condition representing the baseline experience and an ephemeral condition in which temporary, task-scoped components are introduced according to the taxonomy, lifecycle, and design principles defined above. The original condition will present the underlying tasks using a conventional persistent interface, whereas the ephemeral condition will surface temporary interface elements only when a user reaches a context in which additional guidance, refinement, or task support is needed. In this way, the comparison isolates the effect of ephemeral augmentation against the non-ephemeral alternative.

The website presents one scenario that represents a bounded productivity task within the application: a short presentation-refinement exercise foregrounding *refinement support*. Participants receive a small slide deck whose order has been intentionally scrambled and must restore it to a coherent narrative sequence before submitting. The interaction is deliberately narrow: the main slide canvas functions as a large read-only preview, while the participant performs the

task by reordering the slide thumbnails. The lifecycle of trigger, instantiation, interaction, and dismissal is implemented as the common interaction logic for both interface conditions. In the ephemeral condition, temporary support may propose order cues, comparisons, or lightweight explanations of narrative flow, but it does not edit slide content or take over the task. Success is therefore defined as submitting the deck in the predefined canonical order. Figure 3 shows the implemented study interface in both conditions.

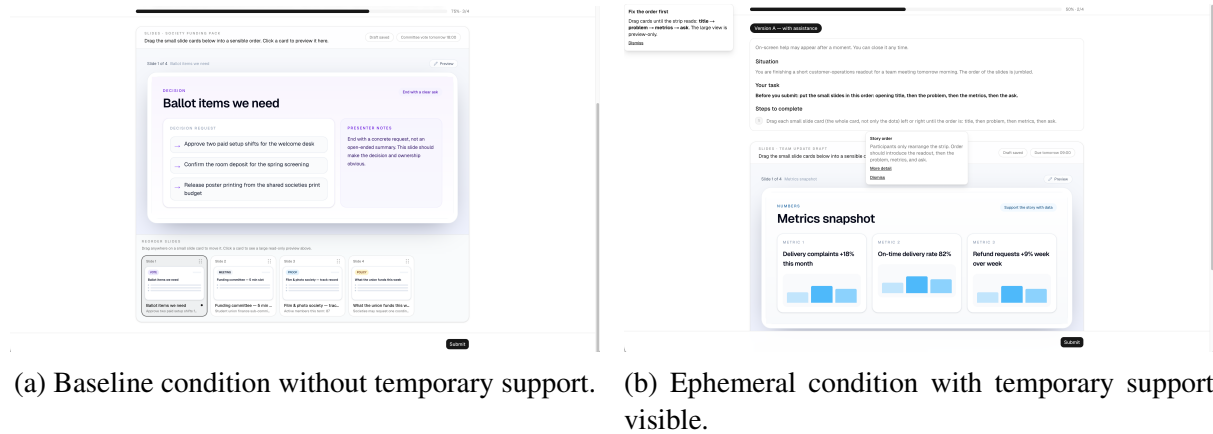


Figure 3: Screenshots of the implemented experiment interface. In both conditions, participants complete the same slide-reordering task: the large slide canvas is a read-only preview, while the actual task action is performed through the thumbnail strip. The ephemeral condition preserves the same underlying task structure but adds temporary, dismissible support overlays.

The ephemeral condition relies on a constrained support-generation pipeline rather than unrestricted interface generation. When support is requested or triggered, the system produces a structured interface specification from the current task context, validates it against schema and safety constraints, and either renders it as a temporary overlay or falls back to a deterministic alternative. The technical details of this pipeline, including the specification model, the validation stages, and the rendering layer, are described in Section 3.

Although the ephemeral condition is allowed to be visually and structurally expressive, that expressiveness remains bounded by the framework. Temporary support may add task-relevant interface elements, annotate or highlight existing content, expose comparison views, or locally reconfigure a small region of the page. However, it does not replace the full application, arbitrarily reorganise the entire interface, or perform consequential actions without confirmation. In practice, the role category, lifecycle, task goals, and success criteria are fixed in advance, while local details such as wording, defaults, and the exact form of low-risk support may vary

in response to immediate context. The ephemeral version is therefore not intended to be wholly different for every participant. Instead, it is constrained adaptation: users encounter the same scenario structures and the same framework rules, but the temporary support may be parameterised slightly differently within those limits. This preserves analytical robustness while still allowing the artifact to reflect the broader motivation of adaptive, goal-relevant ephemeral interfaces.

## **2.4 Evaluation Design**

### **2.4.1 Experimental Design**

The framework will be evaluated through a comparative within-subjects user study using the website artifact described above. Each participant will complete two trials in total: the same slide-refinement scenario once as Version A and once as Version B. The presentation order of those versions is fixed, with Version A always shown first and Version B second, but the mapping between version label and experimental condition is randomised per participant so that one participant may receive the baseline first while another receives the ephemeral condition first under the alternative label. A within-subjects design is appropriate because it allows each participant to act as their own control, thereby reducing the influence of individual differences in digital literacy, task strategy, or prior familiarity with similar interfaces. This version-based counterbalancing preserves a consistent study flow while still varying which version corresponds to assistance across participants.

### **2.4.2 Participants**

Participants will be recruited as regular users of web-based applications who can complete common digital tasks in a desktop browser environment. Recruitment will follow a convenience sampling approach appropriate to the scope of the capstone project. Inclusion criteria will focus on participants who are able to use standard web interfaces independently, while any exclusion criteria will be limited to factors that would prevent valid completion of the study tasks. The final sample size will depend on recruitment feasibility, but the study is designed to support repeated-measures comparisons between the two interface conditions.

### 2.4.3 Materials and Procedure

The primary study material is the experimental website, which implements both the original and ephemeral interface conditions. Participants will complete one bounded scenario twice: a presentation-style refinement task in which they reorder a short slide deck into the intended story sequence. The task structure, goals, and success criteria remain constant for all users so that the study remains comparable across sessions. Within that shared structure, the ephemeral condition may vary in how temporary support is presented so that the assistance can respond to local context while remaining constrained by predefined design rules derived from the framework. In other words, the study does not test unlimited personalisation or unrestricted interface takeover. It tests bounded interface adaptation in which temporary support may add, highlight, compare, or locally rearrange elements while preserving interface continuity and user control.

The procedure will be fully website-based. After opening the study website, participants will read the study information, provide consent, and complete a short background questionnaire about their comfort with websites and familiarity with AI-assisted tools. The website will then present two task trials in a fixed A-then-B sequence, representing the same scenario under the two experimental conditions, with the condition-to-version mapping randomised per participant. Before each trial, participants will receive a short task prompt that labels the interface version for fair debriefing. During each trial, the website will automatically log behavioural data. After each trial, participants will complete brief ratings on intrusiveness, perceived control, and helpfulness. After both trials are complete, participants will answer a final comparative questionnaire about overall preference between the two conditions, with an explicit reminder of which version corresponded to assistance for that participant. This fully instrumented procedure keeps the study structure consistent while allowing all behavioural and self-reported data to be collected in the same environment.

### 2.4.4 Measures

The evaluation combines behavioural and self-reported measures in order to capture both task performance and perceived experience. Behavioural measures are derived from interaction logs recorded by the website, while subjective measures are collected through short post-trial and

post-study questionnaires. The core claim being tested is not simply that ephemeral interfaces are preferred, but that they can improve task support without introducing additional disruption. To assess that claim, the study will measure efficiency through task completion time, effectiveness through task completion rate, friction through interaction count, and user experience through ratings of intrusiveness, perceived control, helpfulness, and overall preference. In addition, ephemeral-only interaction logs will record whether temporary support was triggered, requested, shown, used, dismissed, ignored, or expanded for further detail, allowing the study to interpret how participants actually engaged with the generated assistance.

Table 1: Dependent measures and collection methods.

| Measure              | Type          | Collection             | Analysis                  |
|----------------------|---------------|------------------------|---------------------------|
| Task completion rate | Binary        | Event logging          | McNemar’s exact test      |
| Task completion time | Continuous    | Timestamps             | Wilcoxon signed-rank test |
| Interaction count    | Discrete      | Event logging          | Wilcoxon signed-rank test |
| Intrusiveness        | Ordinal (1–7) | Post-trial Likert item | Wilcoxon signed-rank test |
| Perceived control    | Ordinal (1–7) | Post-trial Likert item | Wilcoxon signed-rank test |
| Helpfulness          | Ordinal (1–7) | Post-trial Likert item | Wilcoxon signed-rank test |
| Preference           | Nominal       | Final forced choice    | Binomial test             |

### 2.4.5 Data Collection

Data collection will combine instrumented event logging with questionnaire responses. For each trial, the website will record task start and end times, completion outcomes, interaction counts, submitted task state, and the active interface condition. In the ephemeral condition, additional logs will capture support trigger events, explicit requests, support shown events, dismissals, usage confirmations, ignored support, detail expansions, and the generated support outputs themselves. Post-trial survey items will capture perceived intrusiveness, perceived control, and helpfulness, while the final questionnaire will capture overall preference between conditions. To link behavioural and survey data, each session will be associated with a participant identifier that is stored consistently across the website logs and questionnaire responses.

### 2.4.6 Analysis Plan

The primary analysis will compare the original and ephemeral conditions on each dependent measure using paired statistical tests selected for the data type, scale of measurement, and expected sample size.

Task completion rate is a paired binary outcome: each participant either completes the task correctly or does not, under each condition. This will be analysed using McNemar's exact test, which evaluates whether the proportion of participants who change completion status between conditions differs from what would be expected by chance. Completion proportions for each condition will be reported alongside the test result.

Task completion time and interaction count are continuous and discrete measures, respectively, that are paired across conditions. Because the expected sample size is modest and time data is typically right-skewed, both measures will be analysed using the Wilcoxon signed-rank test, a non-parametric alternative to the paired t-test that does not assume normality of the paired differences. Medians and interquartile ranges will be reported for each condition.

Post-trial Likert ratings for intrusiveness, perceived control, and helpfulness are ordinal measures on a seven-point scale. Each rating will be analysed using the Wilcoxon signed-rank test, which is appropriate for paired ordinal data. Medians will be reported rather than means to respect the ordinal nature of the scale.

Overall preference is a forced-choice measure in which each participant selects which condition they preferred. This will be analysed using a binomial test against the null hypothesis that preference is split equally between conditions. The count and proportion choosing each condition will be reported.

Ephemeral-only interaction logs will be analysed descriptively to show the proportion of participants for whom support was triggered, shown, used, dismissed, ignored, or expanded, which helps interpret whether a null or positive effect reflects the quality of the support itself or low engagement. The analysis will therefore focus on whether the ephemeral condition yields improvements in efficiency, effectiveness, or perceived helpfulness without increasing intrusiveness or reducing perceived control. Because the ephemeral experience may vary within bounded framework rules, the analysis treats that variation as part of the intended adaptive

behaviour of the artifact rather than as an uncontrolled change to the task itself.

### 3 Implementation

The previous section defined the conceptual framework, the artifact’s purpose as a research instrument, and the evaluation design that structures the comparative study. This section describes how those ideas were translated into a working system. The goal is not to document every engineering decision, but to make the technical design legible enough that the reader can assess whether the artifact faithfully operationalises the framework and whether the instrumentation is sufficient to support the evaluation.

#### 3.1 Architecture Overview

The artifact is built as a full-stack web application using the Next.js App Router with TypeScript. The server layer exposes a set of API routes that handle participant management, trial orchestration, event logging, questionnaire storage, and ephemeral support generation. Persistent data is stored in a PostgreSQL database accessed through Drizzle ORM, which provides type-safe queries and a migration-based schema workflow. The ephemeral support system relies on Google Gemini, accessed through the Vercel AI SDK, to produce structured interface specifications from the current task state. The front end uses Tailwind CSS and a component library based on Radix UI primitives for consistent, accessible interface elements across both experimental conditions.

These choices reflect the requirements of a research instrument rather than a production application. Next.js was selected because its App Router collocates server-side logic and client rendering in the same project, which simplifies the instrumentation of both request handling and user interaction. TypeScript and Zod provide compile-time and runtime type safety across the generation pipeline, which is critical when validating AI-generated output against a constrained specification. PostgreSQL and Drizzle offer relational integrity and migration tooling appropriate for structured experimental data. The Vercel AI SDK abstracts provider-specific details while supporting structured output modes, which allows the generation step to request JSON that conforms to a predefined schema rather than parsing free-form text.



[Diagram placeholder.] Three-tier block diagram: browser client (React, Tailwind, @dnd-kit, ephemeral overlay) → Next.js API (participants, trials, events, support, questionnaires) → PostgreSQL and Google Gemini (structured output). Arrows indicate data flow.

Figure 4: High-level system architecture of the experimental artifact.

### 3.2 Participant Flow and Session Management

The study procedure described in the methodology is implemented as a linear state machine that determines which page the participant should see at any point during the session. When a participant first opens the website, they are presented with the study information and consent form. Providing consent creates a participant record in the database and sets an HTTP-only session cookie that associates all subsequent requests with that record. No traditional authentication is required because the study collects no directly identifying personal data; the cookie serves only to link a browser session to its corresponding participant row.

After consenting, the participant completes a short background questionnaire covering age range, occupation, familiarity with web applications, and familiarity with AI-assisted tools. Submitting the background questionnaire acknowledges the study instructions and triggers the creation of the first trial row. From this point forward, a server-side state resolution function inspects the participant’s record, their trial rows, and their questionnaire responses to determine the current study step. The possible steps form a fixed sequence: *background*, *study* (an active trial), *post-trial questionnaire*, *final questionnaire*, and *complete*. The client-side routing function maps each step to the corresponding page, so that refreshing the browser or navigating away always returns the participant to the correct point in the procedure.

Trial rows are created incrementally rather than all at once. The first trial is created when the participant acknowledges the instructions. Each subsequent trial is created only after the previous trial has been completed and its post-trial questionnaire has been submitted. This ensures that the participant cannot skip ahead or access later trials before finishing earlier ones.

Counterbalancing is implemented at the participant level. When a participant record is created, a boolean flag is randomly set to determine whether the baseline condition corresponds to Version A or Version B. The trial plan then assigns conditions to version labels using this flag, so that one participant may encounter the baseline first while another encounters the ephemeral

condition first, both under the neutral label “Version A.” The content variant for each condition is further diversified by hashing the participant identifier together with the scenario identifier, ensuring that the specific slide deck a participant sees in the baseline condition differs from the one they see in the ephemeral condition.

*[Diagram placeholder.]* Sequential flow: Landing → Consent → Background → Trial 1 → Post-trial 1 → Trial 2 → Post-trial 2 → Final questionnaire → Completion. Annotate `resolveStudyState` and incremental trial creation.

Figure 5: Participant flow through the experimental website. Each step is enforced by a server-side state machine that prevents skipping or reordering.

### 3.3 Data Model

The database schema consists of five tables designed to capture the behavioural and self-reported data required by the evaluation measures defined in the methodology. Figure 6 shows the entity–relationship structure: each participant may complete several trials; each trial may emit many logged events and many stored ephemeral support generations; questionnaire responses belong to a participant, and may optionally reference a trial (post-trial items) or omit a trial key (final questionnaire). Table 2 summarises what each table contributes to the evaluation. The `trial_events` table also stores `participant_id` for efficient querying, in addition to `trial_id`; the diagram omits that redundant edge for clarity.

Table 2: Summary of each table’s role relative to the evaluation measures. Full column definitions follow the Drizzle schema in the artifact.

| Table                                | Role in the study                                                                                                                                   |
|--------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|
| <code>participants</code>            | Consent and background questionnaire; counterbalancing flag ( <code>baseline_is_version_a</code> ); links all rows for a session.                   |
| <code>trials</code>                  | One row per task trial: condition, scenario, timing, correctness, submitted answer, interaction count—supports performance and efficiency measures. |
| <code>trial_events</code>            | Typed events with JSON payloads—supports interaction counts and ephemeral engagement (e.g. support shown, dismissed).                               |
| <code>questionnaire_responses</code> | Post-trial Likert items and final preference; <code>trial_id</code> set for post-trial rows, null for final rows—supports subjective measures.      |
| <code>support_outputs</code>         | Ephemeral condition only: model input state and output spec per generation—supports analysis of support quality and fallback.                       |

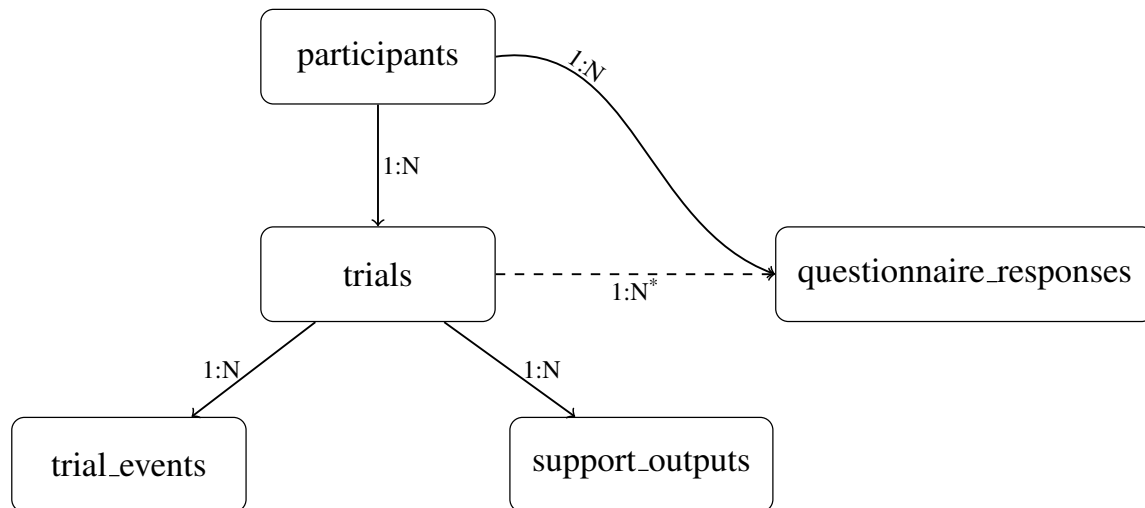


Figure 6: Entity–relationship diagram of the study database. Directed edges are labelled with cardinalities: one participant to many trials; one trial to many events and many support records; one participant to many questionnaire rows. The dashed edge indicates an optional foreign key: `trial_id` is NULL for final-questionnaire rows.

Foreign key constraints with cascading deletion ensure referential integrity: deleting a participant removes all associated trials, events, questionnaire responses, and support outputs. Indexes on participant identifiers, trial identifiers, and event types support efficient querying during data export and analysis.

### 3.4 Task Scenario Implementation

The study evaluates one scenario, the slide-reordering task described in the methodology, under two experimental conditions. The implementation uses a scenario registry that defines each scenario’s metadata, including its taxonomy category, correct answer, allowed ephemeral targets, task prompts, and a function that builds the initial task state. This registry-based design means that adding a new scenario requires only a new registry entry rather than changes to the study flow or state machine.

The slide-reordering scenario presents participants with a short deck of four slides whose order has been intentionally scrambled. The main canvas area displays a read-only preview of the currently selected slide, while the actual task interaction occurs through a horizontal strip of draggable thumbnail cards. Drag-and-drop is implemented using the `@dnd-kit` library, which provides accessible keyboard and pointer-based reordering with sortable constraints.

Participants reorder the strip and submit when they believe the sequence is correct.

Correctness is evaluated by comparing the submitted slide order against a predefined canonical order. Each scenario variant defines its own canonical sequence: variant A follows a corporate readout structure (title, problem, metrics, ask), while variant B follows a committee funding structure (meeting frame, policy, proof, ballot items). The variant assignment uses a deterministic hash of the participant identifier and scenario identifier to assign variant A or variant B to the baseline condition, with the opposite variant assigned to the ephemeral condition. This ensures that each participant encounters two different slide decks across the two trials, preventing familiarity effects from the task content while keeping the task structure identical.

Both conditions share the same task component, the same interaction mechanics, and the same submission logic. The only difference is whether the ephemeral support layer is active, which is controlled by the condition field on the trial row. This design isolates the effect of ephemeral augmentation from other variables.

### 3.5 Ephemeral Support System

The ephemeral support system is the core technical contribution of the artifact. It translates the framework’s concepts of taxonomy, lifecycle, and design principles into a constrained generative pipeline that produces temporary interface overlays without replacing the host application. This subsection describes the three layers of the system: the specification model that defines what can be generated, the pipeline that produces and validates specifications, and the rendering layer that displays them.

#### 3.5.1 Component Specification Model

Ephemeral support is represented as a typed tree structure called an `EphemeralSpec`. Each specification contains a version number, a root node, and metadata indicating whether the overlay is dismissible. Nodes are recursive: each node has a type, a property bag validated against a per-type schema, and optional children. The tree is constrained to a maximum depth of four levels and a maximum of six children per node.

The component catalog defines seventeen component types, organised into five functional

categories. Table 3 lists the types and their roles. Visual cue components draw attention to specific interface elements without adding content. Anchored annotation components attach textual or rich content near a target element. Relational components connect or compare two or more targets to support reasoning across elements. Layout containers position children within the viewport or relative to a target. Rich content components render sanitised HTML and inline SVG for more expressive explanations such as flow diagrams or annotated illustrations.

Table 3: Ephemeral component catalog (version 3). Each type is validated against a per-component Zod schema that enforces field types, string lengths, and structural constraints.

| Category             | Component type    | Role                                                     |
|----------------------|-------------------|----------------------------------------------------------|
| Visual cues          | HighlightRing     | Amber ring around a target element                       |
|                      | PulseRing         | Animated ring with configurable duration                 |
|                      | ArrowCue          | Arrow pointing at a target                               |
|                      | FocusMask         | Dims everything except the target                        |
| Anchored annotations | AnchoredTooltip   | Plain-text tooltip near a target                         |
|                      | HintStack         | Ordered list of short hints near a target                |
|                      | InspectPanel      | Expandable panel with summary and optional detail lines  |
|                      | ConsequenceNote   | Single italic consequence line near a target             |
|                      | SideNote          | Margin annotation beside a target                        |
| Relational           | ConnectorLine     | Dashed line between two targets                          |
|                      | ComparisonStrip   | Short contrast text positioned under two targets         |
|                      | StepRail          | Numbered callouts across two to six targets              |
| Layout containers    | Stack             | Vertical sequence container                              |
|                      | ViewportPanel     | Absolutely positioned panel in viewport percentages      |
|                      | TargetOffsetPanel | Panel positioned relative to a target with pixel offsets |
| Rich content         | FlowHtml          | Sanitised HTML block (only inside ViewportPanel)         |
|                      | AnchoredHtml      | Sanitised HTML anchored near a target                    |

Each component type is associated with a Zod schema that validates its properties at runtime. These schemas enforce constraints such as maximum string lengths, numeric ranges, and required fields. For example, an `InspectPanel` requires a target identifier, a title of at most 120 characters, a summary of at most 280 characters, and optionally up to four detail lines of at most 200 characters each. These limits prevent the model from generating excessively long or unbounded output while leaving enough room for contextually relevant content.

The catalog is versioned so that changes to the available component types or their constraints can be tracked across study sessions. The current study uses catalog version 3.

### 3.5.2 Generation and Validation Pipeline

When ephemeral support is requested during a trial, the system executes a multi-stage pipeline that generates a specification, validates it against the catalog and scenario constraints, and either accepts it or falls back to a deterministic alternative. Figure 7 illustrates this pipeline.

The generation step constructs two prompts. The system prompt describes the specification schema, lists all available component types with their property schemas, states the allowed target identifiers for the current scenario, provides taxonomy-specific guidance (such as which components are most appropriate for refinement support), and enumerates the design rules that constrain output structure. The user prompt includes the current task state, including slide order, deck metadata, and review goals, together with an optional participant interaction snapshot that reflects the participant’s progress at the moment support was requested. This snapshot allows the model to tailor its output to what the participant has already done rather than generating generic first-visit guidance.

The system sends these prompts to Google Gemini using the Vercel AI SDK’s structured output mode, which constrains the model to produce JSON conforming to a provided Zod schema. The schema used for model output is a simplified subset of the full specification schema, designed to avoid compatibility issues with the provider’s structured output implementation while remaining strict enough to prevent most structural errors at generation time.

The model’s output then passes through a series of server-side validation stages:

1. **Envelope parsing.** The top-level structure is checked for a valid version number, root node, and metadata.
2. **Node tree parsing.** The root and its descendants are recursively validated for correct node shape, child count limits, and maximum depth.
3. **Per-component property validation.** Each node’s properties are validated against its type-specific Zod schema, enforcing field types, string lengths, and numeric ranges.

4. **Target allowlisting.** Every target identifier referenced in the specification is checked against the scenario’s predefined list of allowed targets. Any reference to an interface element not declared in the scenario registry causes the specification to be rejected.
5. **Structural container rules.** Layout containers are checked to ensure they contain at least one child. This applies to `Stack`, `ViewportPanel`, and `TargetOffsetPanel`.
6. **Ancestry constraints.** `FlowHtml` nodes are permitted only within a `ViewportPanel` subtree, preventing free-floating rich HTML from appearing anywhere in the interface.
7. **HTML sanitisation.** Any HTML content in `FlowHtml`, `AnchoredHtml`, or `SideNote` nodes is passed through a server-side sanitisation step using `DOMPurify` with a strict allowlist of tags and attributes. This prevents injection of scripts, event handlers, or disallowed elements while permitting semantic HTML and inline SVG for diagrams.

If any stage rejects the output, the system falls back to a deterministic, hardcoded specification for the current scenario. Fallback specifications are defined per scenario and per variant, ensuring that every participant in the ephemeral condition receives some form of support regardless of model behaviour. The fallback specifications use the same component types and target identifiers as model-generated output, so they appear visually consistent with the rest of the ephemeral experience.

Whether the specification was model-generated or a fallback, the full input state, model name, and output are persisted to the `support_outputs` table. This logging enables post-study analysis of generation quality, fallback frequency, and the relationship between input context and output content.

*[Diagram placeholder.]* Expanded pipeline: task state and snapshot → prompts → Gemini structured output → validation (envelope, node tree, props, target allowlist, structural rules, ancestry, HTML sanitisation) → accept or fallback → persist to `support_outputs` → render overlay.

Figure 7: Ephemeral support generation and validation pipeline. Model output passes through seven validation stages before rendering; invalid output triggers a deterministic fallback. Both paths are logged for later analysis.

### 3.5.3 Rendering Layer

Accepted specifications are rendered as a temporary overlay above the existing application interface using a React portal. The overlay component receives the validated specification tree and recursively dispatches each node to its corresponding catalog component based on the node's type. This dispatch model means that the renderer does not need to know about the content of any particular specification; it only needs to map types to components and pass through validated properties.

Components that reference target elements, such as `HighlightRing`, `AnchoredTooltip`, or `ConnectorLine`, resolve their targets by looking up DOM elements by identifier. A custom hook observes the target element's bounding rectangle and updates the overlay position when the layout changes, for example when the browser is resized or when other elements shift. A viewport clamping utility ensures that anchored components remain visible even when their target is near the edge of the screen.

The overlay respects the lifecycle stage of dismissal defined in the framework. When the specification's metadata marks the overlay as dismissible, a close control is rendered that allows the participant to remove the support at any time. Dismissal fires a logged event so that the analysis can distinguish between support that was actively used and support that was dismissed without engagement. The overlay does not persist across page navigations or trial boundaries, which enforces the ephemerality principle at the session level.

## 3.6 Instrumentation and Logging

The evaluation design in the methodology specifies several behavioural and self-reported measures. The artifact implements a corresponding instrumentation layer that captures the data needed for each measure.

Client-side interaction events are sent to a logging API endpoint that inserts rows into the `trial_events` table. Each event includes a type label and a JSON payload with contextual details. A predefined set of event types is recognised as countable interactions, and the trial's running interaction count is incremented automatically when one of these events is recorded.



Table 4 lists the event types and their relationship to the evaluation measures.

Table 4: Logged event types and their relationship to evaluation measures. Events marked with an asterisk (\*) increment the trial’s interaction count.

| Event type                             | When recorded                                                   | Evaluation measure |
|----------------------------------------|-----------------------------------------------------------------|--------------------|
| <code>trial_viewed*</code>             | Participant opens a trial                                       | Interaction count  |
| <code>slide_reordered*</code>          | A slide is dragged to a new position                            | Interaction count  |
| <code>answer_selected*</code>          | An answer is first chosen                                       | Interaction count  |
| <code>answer_changed*</code>           | An answer is revised                                            | Interaction count  |
| <code>support_requested*</code>        | Participant explicitly requests support                         | Support engagement |
| <code>support_triggered*</code>        | Support is triggered automatically                              | Support engagement |
| <code>support_shown*</code>            | Support overlay becomes visible                                 | Support engagement |
| <code>support_dismissed*</code>        | Participant closes the overlay                                  | Support engagement |
| <code>support_used*</code>             | Participant acts on the support                                 | Support engagement |
| <code>support_ignored*</code>          | Support is present but not interacted with                      | Support engagement |
| <code>support_inspect_expanded*</code> | Participant expands an <code>InspectPanel</code> detail section | Support engagement |

For the ephemeral condition, additional data is captured beyond interaction events. Each call to the support generation endpoint persists the full input state (task state and participant snapshot), the model name, and the complete output specification to the `support_outputs` table. This allows the analysis to examine not only whether support was shown and used, but also what the system produced and why, including whether a fallback was triggered.

Questionnaire responses are stored as individual question–response pairs rather than as monolithic form submissions. Post-trial ratings for intrusiveness, perceived control, and helpfulness are linked to the specific trial they follow, while final comparative questions are linked only to the participant. This structure supports direct joins between behavioural data and subjective ratings at the trial level.

A data export endpoint produces a structured bundle of all study data, protected by a server-side secret. The export includes participant records, trial summaries with performance metrics, event logs, questionnaire responses, and support generation outputs, providing the complete dataset needed for the analysis plan described in the methodology.

## 4 Results

This section reports the outcomes of the comparative user study. All measures follow the analysis plan described in the methodology: behavioural data was extracted from the experiment database, and self-reported ratings were collected through post-trial and post-study questionnaires administered within the same website. Unless stated otherwise, paired comparisons use the Wilcoxon signed-rank test, medians and interquartile ranges (IQR) are reported for continuous and ordinal measures, and statistical significance is assessed at  $\alpha = .05$ .

### 4.1 Participants and Data Overview

A total of 0 participants accessed the study website. Of these, 19 completed both task trials and the final questionnaire and are included in the analysis. The remaining participants were excluded because they did not finish both conditions or did not complete the post-study questions.

Table ?? summarises the background characteristics of the included sample.

Table 5: Participant background characteristics ( $n = 19$ ).

| Characteristic                    | Summary                                        |
|-----------------------------------|------------------------------------------------|
| Age range                         | 19 (5); 20 (1); 21 (3); 22 (6); 23 (3); 24 (1) |
| Web application familiarity (1–7) | Mdn = 6.0, IQR = 6.0–7.0                       |
| AI tool familiarity (1–7)         | Mdn = 6.0, IQR = 5.0–7.0                       |
| Condition order: baseline first   | 11 (58%)                                       |
| Condition order: ephemeral first  | 8 (42%)                                        |

### 4.2 Task Completion Rate

Task completion was defined as submitting the slide deck in the correct canonical order. In the baseline condition, 11 of 19 participants (57.9%) completed the task correctly. In the ephemeral condition, 13 of 19 (68.4%) completed correctly. McNemar’s exact test comparing discordant pairs yielded  $p = 0.500$ .

### 4.3 Task Completion Time

Completion time was measured from the moment the participant started the trial to the moment they submitted their answer. Participants who completed the task in the baseline condition had a median time of 35.7 seconds (IQR: 16.4–73.4), compared to 56.9 seconds (IQR: 23.0–85.1) in the ephemeral condition. A Wilcoxon signed-rank test did not yield a statistically significant difference ( $W = 80$ ,  $p = 0.568$ ,  $r = 0.158$ ).

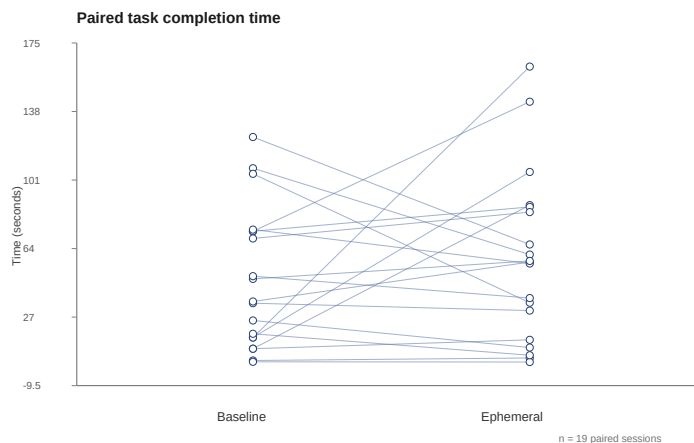


Figure 8: Paired comparison of task completion time (seconds) between the baseline and ephemeral conditions. Each line connects the same participant across conditions.

### 4.4 Interaction Count

The total number of logged interactions per trial was used as a proxy for task effort. The median interaction count in the baseline condition was 3.0 (IQR: 1.0–4.0), compared to 11.0 (IQR: 5.0–13.0) in the ephemeral condition. A Wilcoxon signed-rank test yielded a statistically significant difference ( $W = 0$ ,  $p = <0.001$ ,  $r = 1.000$ ).

### 4.5 Subjective Ratings

After each trial, participants rated the interface on three dimensions using a seven-point Likert scale. Table ?? summarises the results.

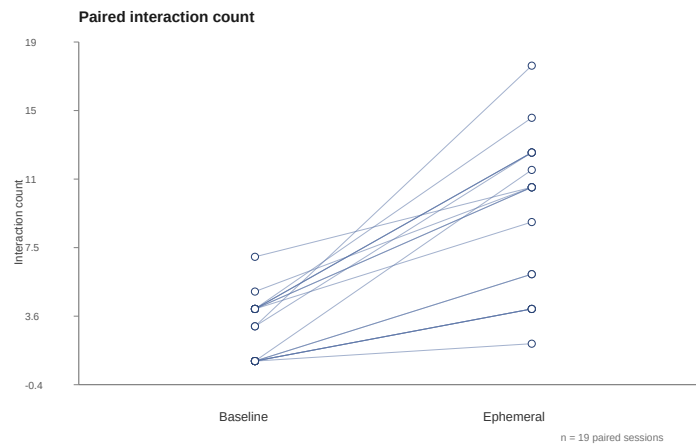


Figure 9: Paired comparison of interaction count between the baseline and ephemeral conditions.

Table 6: Post-trial subjective ratings (7-point Likert scale, 1 = low, 7 = high).

| Measure           | Baseline |         | Ephemeral |         | Wilcoxon<br>$p$ |
|-------------------|----------|---------|-----------|---------|-----------------|
|                   | Mdn      | IQR     | Mdn       | IQR     |                 |
| Helpfulness       | 5.0      | 3.0–6.0 | 6.0       | 5.0–6.0 | 0.048           |
| Intrusiveness     | 3.0      | 2.0–5.0 | 5.0       | 4.0–6.0 | 0.021           |
| Perceived control | 6.0      | 4.5–7.0 | 6.0       | 4.5–6.0 | 0.635           |

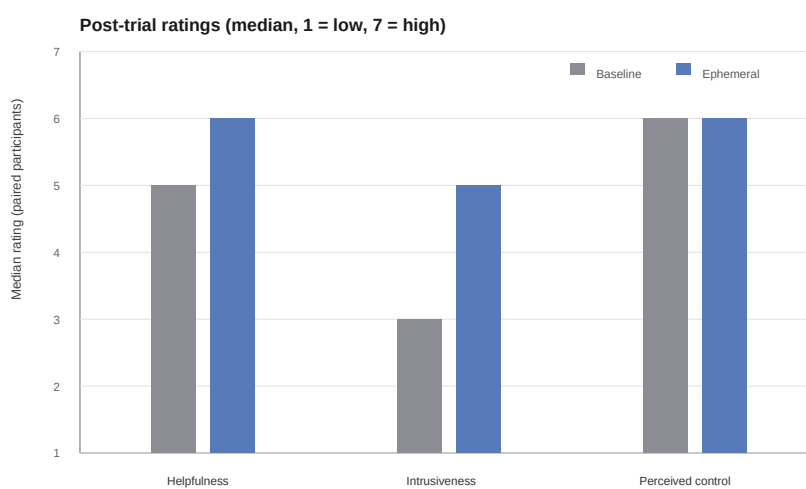


Figure 10: Median post-trial ratings by measure and condition (1 = low, 7 = high). Bars show the median rating across paired participants for each measure.

## 4.6 Overall Preference

After completing both trials, participants answered four forced-choice comparison questions. Responses were stored as Version A or Version B and mapped back to the actual experimental condition using each participant’s counterbalancing assignment. Table ?? summarises the results.

Table 7: Final comparative preference questions ( $n = 19$ ).

| Question             | Baseline | Ephemeral | No preference |
|----------------------|----------|-----------|---------------|
| Overall preference   | 4        | 9         | 6             |
| More helpful         | 6        | 10        | 3             |
| More intrusive       | 1        | 15        | 3             |
| Real-life preference | 7        | 8         | 4             |

## 4.7 Ephemeral Support Engagement

To interpret the behavioural and subjective results in context, this subsection reports how participants actually interacted with the ephemeral support layer. These measures apply only to the ephemeral condition.

Of the 19 participants who completed the ephemeral trial, support was triggered or explicitly requested in 17 cases (89.5%). Once shown, support was actively used (applied or confirmed) in 4 cases, dismissed without use in 17 cases, and ignored entirely in 2 cases. 3 participants expanded the support for further detail. The support generation pipeline produced 37 specifications in total, of which 37 (100.0%) fell back to the deterministic fallback because the generated output failed validation.

## 4.8 Summary of Findings

Table ?? provides an overview of all primary outcome measures.

Table 8: Summary of primary outcome measures across conditions.

| Measure                  | Baseline                                             | Ephemeral | <i>p</i> |
|--------------------------|------------------------------------------------------|-----------|----------|
| Completion rate          | 57.9%                                                | 68.4%     | 0.500    |
| Completion time (Mdn, s) | 35.7                                                 | 56.9      | 0.568    |
| Interaction count (Mdn)  | 3.0                                                  | 11.0      | <0.001   |
| Helpfulness (Mdn)        | 5.0                                                  | 6.0       | 0.048    |
| Intrusiveness (Mdn)      | 3.0                                                  | 5.0       | 0.021    |
| Perceived control (Mdn)  | 6.0                                                  | 6.0       | 0.635    |
| Overall preference       | 4 chose baseline; 9 chose ephemeral; 6 no preference |           | 0.267    |

## 5 Discussion

This study examined whether ephemeral, task-scoped generative UI support can improve a bounded productivity task without disrupting the surrounding interface. The discussion should therefore not reduce the evaluation to the simple question of whether one condition was globally “better” than the other. The more important interpretive issue is whether temporary support can improve the immediate task experience while remaining sufficiently bounded, comprehensible, and non-disruptive within an existing interface. In that sense, the contribution of the project lies not only in the measured outcomes, but also in demonstrating how a design framework for ephemeral support can be translated into a concrete research artifact and evaluated against a persistent baseline.

### 5.1 Interpretation and Relation to Prior Work

This framing positions the study somewhat differently from much of the recent discussion around generative UI. Prior work on generative interfaces has often emphasised adaptability, multimodality, and the capacity of AI systems to tailor interaction structures to user goals across contexts (**Bieniek2024GenerativeAI**). That broader literature is useful for motivating why static interfaces may be insufficient in environments where user needs vary rapidly. At the same time, it also highlights a central design tension: the same adaptive power that makes generative UI attractive can also threaten predictability, continuity, and user control. The present study addresses that tension by evaluating a deliberately narrower alternative. Instead of replacing the interface or restructuring the entire workflow, the artifact constrains generative support to a temporary, local, and task-scoped layer. This is a more modest intervention than the full generative UI vision described in the literature, but it is also closer to the practical reality of augmenting existing productivity software.

The project also relates closely to BISCUIT, which showed that ephemeral UIs can act as an intermediate scaffold between user intent and LLM-generated output in computational notebooks (**Cheng2024Biscuit**). In that study, the temporary interface helped users understand generated code, explore alternatives, and reduce the burden of prompt engineering. The present

work builds on that insight but shifts it into a different domain. Rather than supporting novice programmers in a notebook environment, the artifact evaluates ephemeral support in a bounded productivity scenario where the interface must preserve orientation inside an already recognisable application. This matters because it tests whether the value of temporary scaffolding can extend beyond code generation and into more general productivity-oriented refinement work. Even where the interaction remains simpler than BISCUIT's notebook setting, the study supports the broader idea that ephemeral support can be useful when it helps users inspect, steer, or complete an immediate subtask without requiring them to hand over the whole workflow to the system.

The discussion also connects to the earlier concept of ephemeral user interfaces proposed by Döring et al., who argue that intentionally temporary interface elements can reduce overload by limiting persistent accumulation and by presenting information only for the period in which it is relevant (**Döring2013EphemeralUI**). Their work was grounded primarily in material and interaction-design perspectives rather than in web-based productivity systems, yet its core logic remains relevant here. The present thesis translates that logic into a digital software context by treating ephemerality not as a material property, but as a design constraint on interface behaviour. In this formulation, the important question is not whether an element literally disappears because of a physical material, but whether the system introduces assistance only when contextually justified, keeps it bounded to a local task, and lets it recede without destabilising the surrounding workflow. This reinterpretation is one of the conceptual contributions of the project.

Taken together, these comparisons suggest that the main value of the study is not to claim a final answer on generative UI in productivity tools, but to occupy an important middle ground in the literature. On one side are static interfaces that preserve continuity but may struggle to offer flexible, situation-specific assistance. On the other side are highly adaptive or fully generated interfaces that promise responsiveness but risk undermining user orientation and trust. The framework evaluated in this thesis proposes that ephemeral, task-scoped support may offer a compromise between these extremes: assistance can be adaptive, but only within predefined boundaries concerning trigger conditions, scope, continuity, user control, and dismissal.



Whether the measured effects are strongly positive, mixed, or only modest, that middle-ground framing remains analytically useful because it clarifies what exactly is being tested and why it differs from both fully static and fully generative alternatives.

## 5.2 Limitations and Robustness

An important boundary of the study is the scope of the evaluation itself. The artifact was evaluated through a single presentation-refinement scenario that foregrounded refinement support rather than the full range of support roles proposed in the framework. This choice was appropriate for the capstone because it kept the procedure manageable and reduced participant fatigue. A design with three scenarios across two interface conditions would have required six task trials per participant, which would likely have increased drop-off, reduced engagement, and weakened the quality of responses. However, the consequence of this decision is that the findings cannot yet be generalised across multiple task types, domains, or support roles such as interpretive support or task-execution support.

The study is also limited by the breadth of interactions represented in the artifact. Some richer or more open-ended forms of support, such as more advanced writing assistance or more expressive temporary interaction patterns, were not retained in the final evaluation design. This means that the study primarily evaluates a bounded form of ephemeral support in a structured workflow rather than the broader design space of adaptive generative interfaces. As a result, the conclusions are strongest for short, goal-directed refinement tasks and should not be treated as evidence that the same interaction logic would perform equally well in longer writing workflows, highly creative tasks, or more complex multi-step activities.

Further limitations concern sampling, device conditions, and duration. The study relies on a modest participant sample recruited within the practical constraints of the project, which limits the diversity of backgrounds, habits, and expectations represented in the data. In addition, the experimental artifact was not optimised for mobile use. If some participants accessed the study on smaller screens or in device conditions for which the interface was not well adapted, this may have introduced additional friction unrelated to the experimental condition itself and may therefore have skewed measures such as completion time, interaction effort, perceived intru-

siveness, or overall preference. Participants also interacted with the system only within a short study session. This makes the evaluation suitable for examining immediate task performance and first impressions, but it does not capture longer-term effects such as habituation, changes in trust, or whether ephemeral assistance remains helpful once its novelty has worn off. The conclusions should therefore be read as evidence from a controlled short-term evaluation rather than as proof of long-term adoption in real-world settings.

Another limitation is that the framework is tested through one particular implementation of the artifact rather than through multiple independent implementations. Positive or negative outcomes may therefore reflect not only the underlying framework, but also specific design decisions in this prototype, such as how support was phrased, when it was triggered, and how interaction options were presented. Subjective measures such as helpfulness, perceived control, and intrusiveness also rely on participant self-report and are therefore shaped by individual interpretation. These issues do not invalidate the findings, but they do mean that the results should be understood as an evaluation of a concrete operationalisation of the framework rather than a definitive test of ephemeral support in the abstract.

At the same time, several aspects of the research design strengthen the robustness of the study within these boundaries. Both interface conditions were evaluated on the same underlying task with the same success criteria, which supports a fair comparison between the baseline and ephemeral versions. The within-subjects design reduced the influence of individual differences because each participant encountered both conditions, while randomising trial order helped limit systematic ordering effects. The website also combined automated event logging with post-trial questionnaires, making it possible to compare behavioural outcomes and subjective experience rather than relying on anecdotes alone. Finally, although the ephemeral condition allowed bounded adaptation, that variation was constrained by predefined framework rules concerning scope, lifecycle, and user control, which preserved analytical comparability across sessions.

Taken together, these limitations and robustness measures suggest that the study should be interpreted as a focused but credible first evaluation of ephemeral, task-scoped support in productivity software. The results are sufficient to discuss the promise of the framework in a

controlled setting, but they do not establish broad generalisability across contexts. Section ?? sets out concrete directions for extending scenarios, evaluation methods, and the design space of ephemeral components.

## 6 Future Work

The present evaluation is intentionally narrow: one scenario, one operationalisation of the framework, and a remote, self-paced procedure. The directions below follow from that scope and from what the artifact already suggests about where ephemeral, task-scoped support may be most informative next.

### 6.1 Broader scenarios and user populations

The study used a single presentation-refinement workflow. Ephemeral support is unlikely to matter equally in every domain. In software development, for example, many users already rely on mature patterns such as inline completion and tab-style acceptance flows in editors; in that setting, a bounded ephemeral layer may add less distinct value, or may need to compete with tooling that is already deeply integrated into the task. By contrast, everyday office or administrative work—where users must complete structured tasks inside interfaces they use only occasionally—may leave more room for lightweight, task-scoped assistance that does not assume expert shortcuts or a rich existing toolchain. A useful next step is therefore to evaluate the framework across several workflows and to sample participants with different levels of familiarity with the host application and with productivity software in general. Such work would test whether benefits cluster in contexts where users are less able to self-scaffold, and would clarify when ephemeral support is complementary rather than redundant relative to established interaction models.

### 6.2 Deeper evaluation methods

Increasing the number of remote completions would improve statistical power, but it would not on its own address every open question about *why* participants behaved as they did, or how they interpreted ephemeral support moment to moment. A complementary direction is structured, in-depth evaluation: for example moderated sessions with think-aloud protocols, where a researcher can observe navigation problems, misunderstanding of triggers, and emotional reactions that web logs and post-task questionnaires only partially capture. Combining a controlled

quantitative comparison with a smaller set of rich qualitative sessions would better connect design choices to observed friction and to subjective measures such as perceived control and intrusiveness.

### **6.3 Richer ephemeral component catalogues**

The evaluation artifact deliberately used a constrained set of ephemeral UI components so that conditions remained comparable and the study remained feasible within the capstone timeline. The underlying framework could support a wider catalogue: more varied layouts, stronger personalisation of content and tone, and more expressive temporary interaction patterns than were deployed here. Expanding that catalogue raises its own research questions—how to preserve predictability and user agency when surface behaviour varies more widely, and how to evaluate “wilder” or more personalised variants without confounding condition effects with implementation novelty. Future work could treat catalogue design as a first-class object of study: progressive disclosure of complexity, user-tunable boundaries for ephemerality, and systematic comparison of a small number of high-diversity implementations against a stable baseline.

## **7 Conclusion**

This is your conclusions

Your supplementary materials are placed here.